Chair of Mathematical Statistics
Department of Mathematics
Technical University of Munich

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Introduction to Data Science and Statistical Thinking

Exam:	CIT5130002 / Final	**Date:**	Tuesday 23rd July, 2024
Examiner:	Prof. Mathias Drton	**Time:**	08:30 – 10:00

Working instructions

- This exam consists of **14 pages** with a total of **8 problems**.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 34 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **scientific pocket calculator**
  - the **cheat sheet**, which is contained in this exam on page **12**
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Additional space for solutions to **any** of the problems is given on page 13 and 14.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from ________ to ________ / Early submission at ________

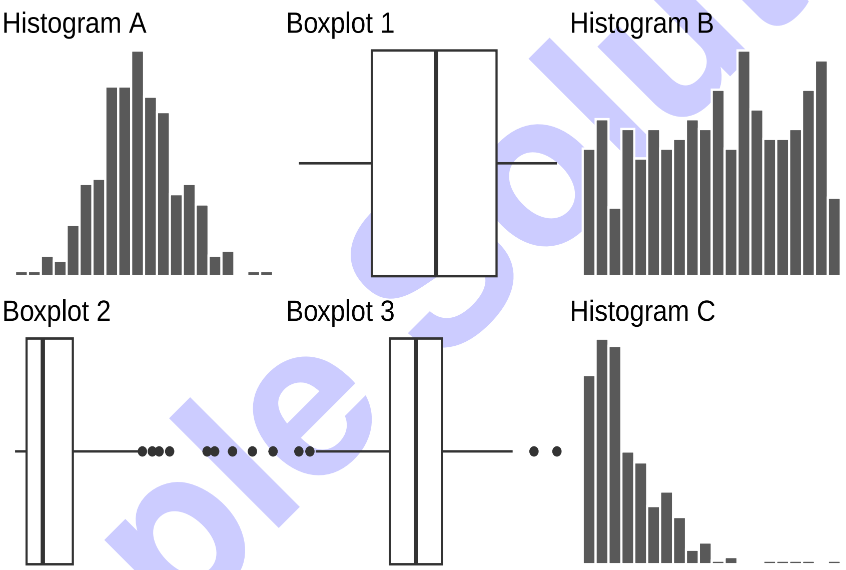
Problem 1 (4 credits)

Give a short answer to each of the following **unrelated** sub-problems.

a) A political scientist is interested in the effect of government type on economic development. To conduct her analysis, she wants to use a sample of 30 countries evenly represented among North America, South America, Europe, Asia, and Africa. What type of study and sampling strategy should she use to ensure that countries are selected from each region?
Hint: Your answer should include at least two of the terms: observational, experimental, simple random sample, cluster sample, stratified sample.

She will only be able to conduct an **observational study**. The five continents represent five different strata and from each stratum a random sample should be drawn. Hence, it should be a **stratified sample**.

b) The figure below displays the histogram and boxplot of **three different** datasets. Match the histogram and boxplot, which belong to the same dataset. You do not need to justify your **choice**.



The correct combinations are: A-3, B-1 and C-2.

c) A professor graded a test for 140 students and found that the median number of points was 39. Upon reviewing the results, she discovered that the student with the lowest score, who originally received three points, should have received only one point due to an error in one of the subtasks. Determine the median of the adjusted test results. Justify your answer.

The empirical median of a sample is any value such that at least half of the sample is less than or equal to the proposed median and at least half is greater than or equal to the proposed median. The score of the lowest performing student was the minimum before she adjusted the score, and it is still the minimum after the adjustment (all other scores are higher). Therefore, the number of points where at least half of the sample is less than or equal to remains unchanged. So, the median will also be 39.

d) After completing an introductory statistics course, 70% of students are able to successfully construct box plots. Among those who can construct box plots, 95% passed the course, while only 52% of those who couldn't construct box plots passed. Calculate the probability that a student is able to construct a box plot if it is known that they passed.

You can either draw a tree or use Bayes formula. The solution uses the latter approach. Let CB and PC be the events that a student can construct a box plot and passes the course, respectively. The probability is then given by:

$$\begin{aligned} \mathrm{P}(CB|PC) &= \frac{\mathrm{P}(PC \cap CB)}{\mathrm{P}(PC)} = \frac{0.7 \cdot 0.95}{0.7 \cdot 0.95 + 0.3 \cdot 0.52} \\ &= \frac{0.665}{0.665 + 0.156} = 0.8099878 \in (0.8, 0.81) \end{aligned}$$

Problem 2 (5 credits)

a) You are interested in probability distributions over the sample space $S = \{1, 2, 3, 4, 5\}$. Proposed probability distributions are P_a, P_b and P_c. They have the following properties:

i) $\mathrm{P}_a(\{x\}) = 0$ for $x \in \{1, 2, 4, 5\}$, and $\mathrm{P}_a(\{3\}) = 1$;

ii) $\mathrm{P}_b(\{1\}) = 0.1$, $\mathrm{P}_b(\{2\}) = \mathrm{P}_b(\{3\}) = \mathrm{P}_b(\{4\}) = 0.3$, $\mathrm{P}_b(\{5\}) = 0.1$;

iii) $\mathrm{P}_c(\{3\}) = 0.2$, $\mathrm{P}_c(\{2, 3\}) = 0.3 = \mathrm{P}_c(\{3, 4\})$, and $\mathrm{P}_c(\{2, 3, 4\}) = 0.6$.

Identify each invalid probability distribution, and explain your reasoning. For each valid probability distribution decide if the events $A = \{2, 3\}$ and $B = \{3, 4\}$ are independent.

i) P_a is a valid probability distribution. The events A and B are independent, if $\mathrm{P}_a(A \cap B) = \mathrm{P}_a(A) \cdot \mathrm{P}_a(B)$, which is the case since

$$\begin{aligned} \mathrm{P}_a(A \cap B) = \mathrm{P}_a(\{3\}) = 1 &= \mathrm{P}_a(\{3\}) \cdot \mathrm{P}_a(\{3\}) \\ &= (\mathrm{P}_a(\{2\}) + \mathrm{P}_a(\{3\})) \cdot (\mathrm{P}_a(\{3\}) + \mathrm{P}_a(\{4\})) \,. \\ &= \mathrm{P}_a(A) \cdot \mathrm{P}_a(B) \end{aligned}$$

ii) P_b is invalid. The sum of the probabilities is greater than 1 ($0.1 + 0.3 + 0.3 + 0.3 + 0.1 = 1.1$).

iii) P_c invalid. For each probability distribution P it must hold $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$. But for $A = \{2, 3\}$ and $B = \{3, 4\}$ we get

$$\begin{aligned} \mathrm{P}_c(\{2, 3\} \cup \{3, 4\}) &= \mathrm{P}_c(\{2, 3, 4\}) = 0.6 \\ &\neq \mathrm{P}_c(\{2, 3\}) + \mathrm{P}_c(\{3, 4\}) - \mathrm{P}_c(\{3\}) = 0.3 + 0.3 - 0.2 \,. \end{aligned}$$

0 1 2

b) P_d is a valid probability distribution over $S = \{1, 2, 3, 4, 5\}$ with $\mathrm{P}_d(\{1\}) = \frac{3}{10}$, $\mathrm{P}_d(\{2\}) = \frac{2}{10}$, $\mathrm{P}_d(\{3\}) = \frac{1}{10}$ and $\mathrm{P}_d(\{4\}) = \frac{2}{10} = \mathrm{P}(\{5\})$. In a game of chance, a number is randomly drawn from the set S based on P_d. You win 1 Euro if a 1 is drawn, 5 Euros if a 2 is drawn, 10 Euros if a 3 is drawn and nothing otherwise. To play this game, you have to pay 3 Euros. What is your expected profit/loss from playing this game?

The profit/loss for playing this game is described through the following random variable:

$$X(s) = \begin{cases} 1 - 3 = -2, & s = 1 \\ 5 - 3 = 2, & s = 2 \\ 10 - 3 = 7, & s = 3 \\ 0 - 3 = -3 & s \in \{4, 5\} \end{cases} .$$

The expected profit/loss from playing this game is then

$$\begin{aligned} \mathrm{E}[X] &= -2 \cdot \mathrm{P}(X = -2) + 2 \cdot \mathrm{P}(X = 2) + 7 \cdot \mathrm{P}(X = 7) + (-3) \cdot \mathrm{P}(X = -3) \\ &= -2 \cdot \mathrm{P}_d(\{1\}) + 2 \cdot \mathrm{P}_d(\{2\}) + 7 \cdot \mathrm{P}_d(\{3\}) - 3 \cdot \mathrm{P}_d(\{4, 5\}) \\ &= -2 \cdot \frac{3}{10} + 2 \cdot \frac{2}{10} + 7 \cdot \frac{1}{10} - 3 \cdot \frac{4}{10} \\ &= -\frac{7}{10} . \end{aligned}$$

Problem 3 (2 credits)

0 1 2

The following training data, containing six observations on three predictors and one qualitative response variable, is given.

```
train

  x1 x2 x3      y
1  0  3  0   blue
2  2  0  0   blue
3  0  1  3   blue
4  0  1  2 yellow
5 -1  0  1 yellow
6  1  1  1   blue
```

Suppose we wish to use this dataset to predict a value for the response Y given the new observation $(x_1, x_2, x_3) = (0, 1, 1)$ using k-nearest neighbors. What is our prediction for $k = 3$? Justify your answer.

Hint:

```
train |>
  mutate(
    dist =sqrt((x1-0)^2+(x2-1)^2+(x3-1)^2))

  x1 x2 x3      y     dist
1  0  3  0   blue 2.236068
2  2  0  0   blue 2.449490
3  0  1  3   blue 2.000000
4  0  1  2 yellow 1.000000
5 -1  0  1 yellow 1.414214
6  1  1  1   blue 1.000000
```

For $k = 3$, the prediction will be `yellow` since the three nearest neighbors of (0,1,1) are (1,1,1), (0,1,2) and (-1,0,1), where `yellow` wins the majority votewith two occurrences.

Problem 4 (5 credits)

US Department of Health and Human Services, Centers for Disease Control and Prevention collect information on births recorded in the country. The data used here are a random sample of 1000 births from 2014. We use logistic regression to model whether the baby is premature, `premie` with levels `1` ("yes") and `0` ("no"), from various predictor variables, like `weight` (weight of the baby at birth in pounds) or `mature` (maturity status of the mother with levels `mature mom` and `younger mom`).

```
model_2p <- glm(premie ~ weight + mature, family = binomial, data = births14)
tidy(model_2p)
```

```
# A tibble: 3 x 5
  term              estimate std.error statistic  p.value
  <chr>                <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)          6.59      0.748      8.80 1.34e-18
2 weight              -1.22      0.108    -11.3  1.65e-29
3 matureyounger mom   -0.570     0.295     -1.94 5.30e- 2
```

a) Give an interpretation of the estimated slope of `mature` in terms of the odds ratio.

The odds ratio of having a premature baby is equal to $e^{-0.57} = 0.5655254$, i.e. that the odds of having a premature baby is decreased by almost 43% for a younger mom compared to a mature mom, given all else being equal.

b) The model should be used to predict whether the baby is premature or not. To choose a suitable threshold, the dataset was split into a training (`births14_train`) and test (`births14_test`) dataset. The model was then fitted to the training data and probabilities for having a premature baby were predicted for the test data.

```
model_2p_train <- glm(premie ~ weight + mature, family = binomial, data = births14_train)
births14_test <- births14_test |>
  add_predictions(model_2p_train, type = "response")
```

The following candidates are considered as threshold values.

```
thresholds<-c(0.4, 0.25, 0.15, 0.1)
```

Which threshold value should be chosen based on the following output. Justify your answer.

```
births14_test |>
  probably::threshold_perf(premie, pred, thresholds, event_level = "second")
```

```
# A tibble: 12 x 4
   .threshold .metric     .estimator .estimate
        <dbl> <chr>       <chr>          <dbl>
 1       0.1  sensitivity binary         0.714
 2       0.15 sensitivity binary         0.714
 3       0.25 sensitivity binary         0.5
 4       0.4  sensitivity binary         0.429
 5       0.1  specificity binary         0.709
 6       0.15 specificity binary         0.814
 7       0.25 specificity binary         0.884
 8       0.4  specificity binary         0.965
 9       0.1  j_index     binary         0.424
10       0.15 j_index     binary         0.528
11       0.25 j_index     binary         0.384
12       0.4  j_index     binary         0.394
```

By choosing the threshold in an optimal way, we would like to maximize sensitivity and specificity. Since we can't optimize both at the same time, we have to find a trade-off, which is described through Youden's J index

$$J = \text{sensitivity} + \text{specificity} - 1.$$

By analyzing the output, we can see that the largest J index occurs when the threshold is 0.15.
Remark: One may also simply consider the sum of sensitivity and specificity, which yields the same result.

0 1 2

c) You ask yourself if you can improve the predictive accuracy by adding additional predictor variables, such as the mother's age (`mage`), the number of hospital `visits` during pregnancy, the weight `gained` by the mother and whether the mother was a smoker or not (`habit`). You therefore apply 10-fold cross-validation on the current model and a second one, which is defined through the following formula:

```
premie ~ weight + mature + mage + visits + gained + habit
```

Which model should you choose based on the following output, the one with two (`fit_folds_2p`) or the one with six (`fit_folds_6p`) predictors. Justify your answer, taking into account both provided summary statistics.

```
collect_metrics(fit_folds_2p, type = "wide")

# A tibble: 1 x 2
  accuracy roc_auc
     <dbl>   <dbl>
1    0.911   0.830

collect_metrics(fit_folds_6p, type = "wide")
# A tibble: 1 x 2
  accuracy roc_auc
     <dbl>   <dbl>
1    0.913   0.837
```

In the output, we discover the average accuracy and AUC value for both models. Higher values for accuracy and the area under the curve indicate greater predictive accuracy. The model using 6 predictor variables shows higher values for accuracy and AUC. Hence, we should choose `model_6p`.

Problem 5 (3 credits)

A large farm is interested in testing a new type of fertilizer to see if it can increase the farm's corn production. The land is divided into plots, with each plot producing an average of 1215 pounds of corn and having a standard deviation of 94 pounds. The owner is interested in detecting any average difference of at least 40 pounds per plot. As a result, she is planning to carry out an experiment to evaluate the effectiveness of the new fertilizer in comparison to the currently used fertilizer. She used R and produced the following result:

```
power.t.test(delta = 40, sd = 94, sig.level = 0.05, power = 0.9,
             type = "two.sample", alternative = "one.sided")

     Two-sample t test power calculation

              n = 95.27179
          delta = 40
             sd = 94
      sig.level = 0.05
          power = 0.9
    alternative = one.sided

NOTE: n is number in *each* group
```

a) Which sample size is needed due to the above given output? Justify your answer. 0 1 2

The output says that she needs 95.27179 observations per group.Since there are two groups, she needs twice as many. Hence, the sample size must be at least $2 \cdot 96 = 192$.

b) Assume she uses the correct sample size. What is the probability of detecting that the new fertilizer results in an increase of 40 pounds or more in the average amount of corn produced per plot, if the fertilizer actually leads to an increase of 40 pounds or more and the standard deviation is equal to 94? Explain your answer. 0 1

The probability of detecting that the new fertilizer results in an increase of 40 pounds or more in the average amount of corn produced per plot, if the fertilizer actually leads to an increase of 40 pounds or more, is the power of this test. Hence, the probability is equal to 0.9.

Problem 6 (2 credits)

A psychologist is aiming to establish whether socio-economic status is associated with preferences for playing games. Fifty children in total were selected from families of low, middle, and high socio-economic status. The children were then asked to choose one of the following games: Monopoly, Battleship, or Connect Four.

a) State the null hypothesis H_0, which the psychologist should use to test her assumptions. 0 1

The psychologist wants to test if the socio-economic status is associated with preferences for playing games. Hence, the null hypothesis must assume that they are not associated or in other words:
H_0 : socio-economic status and game playing preferences are independent.

b) The psychologist has requested your assistance with the statistical analysis. You have chosen to employ a simulation-based approach to test the null hypothesis described in a). Describe how you plan to generate new samples in order to approximate the null distribution of a suitable test statistic.

Under the null hypothesis, the two variables are considered independent. In other words, the preference for playing games is not associated with the socio-economic status. Hence, we can generate new samples by randomly rearranging the values of one variable while keeping all the values of the other variable constant.This allows us to randomly allocate a game to a child, which are characterised through their socio-economic status, or vice versa.

Problem 7 (5 credits)

After graduating from TUM, you decide to go to Monte Carlo for vacation. You visit a casino and choose to gamble, specifically opting to play roulette. Before beginning to place bets, you observe 500 roulette games at the casino. Afterward, you notice that the ball lands on red 260 times. Can you determine if the roulette is fair in terms of showing red?

Hint: In Monte Carlo, roulette involves spinning a wheel with various colored slots, including 1 gold, 18 red, and 18 black.

a) Define the null and alternative hypothesis for testing if the roulette is not fair in terms of showing red.

Let θ be the probability of showing "red". The roulette is fair, if $\theta = \frac{18}{37}$. Hence, the testing problem to consider is given by

$$H_0 : \theta = \frac{18}{37} \qquad \theta \neq \frac{18}{37}.$$

b) To test the hypothesis in a) you use your laptop. In R you create the data frame:

```
df<-data.frame(outcome = c(rep("red", 260), rep("not red", 340)))
```

Now you want to use a simulation-based approach to test if the roulette is fair. You want to use the following code to simulate (suitable) test statistic values under the null hypothesis. Unfortunately, the code is incomplete. Complete the code so that the output `null_distn` can be used as shown in part c).

```
null_distn <- df |>
  specify(
    response = outcome,
    ___________________
  ) |>
hypothesize(
  null = "point",
  ____________
  ) |>
generate(
  reps = 1000,
  type = __________
  ) |>
calculate(
  ________________
  )
```

specify(response = outcome, success = "red")

hypothesize(null = "point", p = 18/37)

generate(reps = 1000, type = "draw")

calculate(stat = "prop")

c) Calculate the p-value using the following output

```
null_distn |>
  filter(stat >= 260/500 | stat <= 18/37 - (260/500 - 18/37)) |>
  summarize(n = n())

# A tibble: 1 x 1
      n
  <int>
1   112
```

Use the p-value to test the null hypothesis at the 0.05 significance level and state the test's conclusion in the data context. Justify your answer.

From the output we can detect that 112 out of 1000 samples were as extreme as the original sample or even more favorable in terms of the alternative. But this means that the p-value is equal to

$$\text{p-value} = \frac{112}{1000} = 0.112,$$

which is larger than the chosen significance level 0.05. Hence, we fail to reject the null hypothesis. The data do not show enough evidence to reject the assumption that the roulette wheel is fair in terms of showing red.

Problem 8 (8 credits)

The dataset `fat` contains 251 measurements of 14 variables, which can be used to predict the percentage of body fat (`body.fat`). Measuring body fat percentage is relatively expensive compared to measuring the other variables. One would have found a cost-effective alternative to measuring the body fat percentage if a good model fit can be achieved. A model using all predictor variables is fitted with the following command.

```
model.full<-lm(body.fat ~ ., data = fat)
```

a) The full model contains uninformative predictor variables, which we want to identify by running a backward selection. In one of the steps, we obtain the following values

```
Step:  AIC=696.23
body.fat ~ age + weight + BMI + neck + chest +
    abdomen + hip + thigh + forearm + wrist

          Df Sum of Sq    RSS    AIC
- age      1     85.80 3769.3 700.01
- weight   1     24.92 3708.4 695.92
<none>                 3683.5 696.23
- hip      1     27.73 3711.2 696.11
- thigh    1     37.58 3721.1 696.77
- wrist    1    172.90 3856.4 705.74
- forearm  1     40.46 3724.0 696.97
- chest    1     16.01 3699.5 695.32
- neck     1     47.42 3730.9 697.44
- BMI      1     50.68 3734.2 697.66
- abdomen  1   1412.08 5095.6 775.68
```

Which variable should be removed (if any) in this step? Justify your answer.
Remark: Be aware that the output is not given in the standard form.

In each step we remove one of the predictor variables, if this leads to a decrease in the AIC value. We pick the variable which results in the largest decline.The AIC value of the current model is 696.23. Removing `chest` leads to an AIC value of 695.32, which is the smallest value given in the output. Hence, `chest` will be removed in this step.

0 1 2 3

b) `model.step` is the model obtained by backward selection. It contains 10 parameters (one intercept and nine slope parameters). For this model the following table contains some part of the summary.

```
# A tibble: 5 x 5
  term        estimate std.error statistic p.value
  <chr>          <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)   3.00      6.86       0.438      NA
2 age           0.0748    0.0273     NA         NA
3 BMI           0.513     0.232      NA         NA
4 neck         -0.426     0.205      NA         NA
5 thigh         0.170     0.122      NA         NA
```

Decide if `age` is significant at the significance level 0.05. Justify your answer by judging the size of the p-value (the values given in the hint allow you to do that). Give an interpretation of the slope estimate for `age`.
Hint:

```
pt(c(1.651, 1.969), df = 241, lower.tail = FALSE)
[1] 0.050 0.025
```

From the hint we know that a statistic value of 1.969 would lead to a p-value of $2 \cdot 0.025 = 0.05$.
To decide if `age` is significant, we need to compute the statistic value

$$T(y, \mathbf{x}) = \frac{0.0748}{0.0273} = 2.7399267 \approx 2.74 \in (2.7, 2.8)\,.$$

Since $2.74 > 1.969$, it follows that

$$P_{H_0}(|T(\mathbf{Y}, \mathbf{x})| > 2.74) < P_{H_0}(|T(\mathbf{Y}, \mathbf{x})| > 1.969) = 2 \cdot P_{H_0}(T(\mathbf{Y}, \mathbf{x})) > 1.969) = 0.05\,.$$

Hence, we can reject the null hypothesis of no effect from `age` at the significance level 0.05, since the p-value is smaller than the significance level.
Given all else equal, the percentage of body fat will increase on average by 0.0748for each additional year of age.

0 1 2

c) Use the table from b) to compute a 95% confidence interval for the slope parameter of `neck`. Use the computed interval to decide if the data provides evidence for a relationship between `body.fat` and `neck` Justify your answer. **Hint:**

```
qt(c(0.95,0.975),df =241)
[1] 1.651201 1.969856
```

The 95% confidence interval for β_{neck} is given by

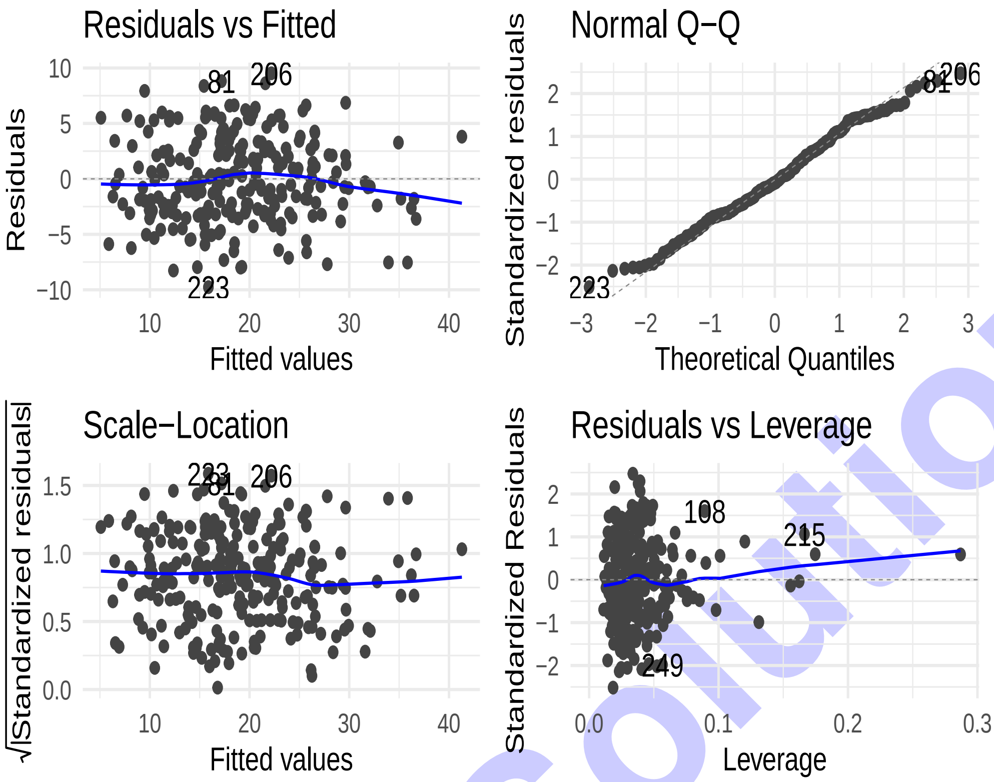
$$\widehat{\beta}_{\text{neck}} \pm t(251-9-1)_{1-0.05/2} \cdot \widehat{\text{SE}}_{\widehat{\beta}_{\text{neck}}} \approx -0.426 \pm 1.96 \cdot 0.205 = (-0.8278, -0.0242)\,.$$

The interval doesn't cover zero. The variables would not be related if the slope parameter were zero. Therfore, the data provides evidence for a relationship between `body.fat` and `neck`.

d) Conduct a residual analysis to check the assumptions of a linear model.

0 1 2

```
autoplot(model.step)
```



The Residuals vs Fitted plot shows that there is (almost) no structure left in the residuals. Since the points fall along the dotted line in the Normal Q-Q plot, we can conclude that the (standardised) residuals are nearly normal. The Scale-Location plot shows that the assumption about a homogeneous variance is probably full-filled, since the blue line more or less a straight line around 1. In the Residuals vs Leverage plot, we observe one observation (around 0.3) with a more considerable leverage value, but the corresponding residual is small. So, the observation is not influential.

Conclusion: The assumptions of the linear model seem to be full-filled. For one observation the model doesn't seem to fit that good. But probably the observation will not be very influential, since the Cook's Distance value is less than 0.5.

Cheat sheet

Explore categorical data:

$f_{j,\ell} = \sum_{i=1}^{n} \mathbf{1}_{(v_j,w_\ell)}\big((x_i, y_i)\big), \quad j \in \{1, \dots, k\}, \ell \in \{1, \dots, m\};$

$f_j = \sum_{\ell=1}^{m} f_{j,\ell};$ $\quad f_\ell = \sum_{j=1}^{k} f_{j,\ell};$ $\quad r_j = \frac{f_j}{n};$

Explore numerical data:

$\overline{x}_n = \frac{x_1+x_2+\cdots+x_n}{n};$

$s_{x,n}^2 = \frac{1}{n-1}\sum_{i=1}^{n} x_i^2 - \frac{n}{n-1}\overline{x}_n^2;$ $\quad IQR = Q_3 - Q_1;$ $\quad Q_1 - 1.5 \cdot IQR;$ $\quad Q_3 + 1.5 \cdot IQR;$

$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x}_n)(y_i - \overline{y}_n);$ $\quad r_{(x,y),n} = \frac{\frac{1}{n-1}\sum_{i=1}^{n} x_i y_i - \frac{n}{n-1}\overline{x}_n\overline{y}_n}{s_{x,n}s_{y,n}};$

Probability:

$\mathrm{P}(A^c) = 1 - \mathrm{P}(A);$ $\quad \mathrm{P}(A \cap B) = \mathrm{P}(A) \cdot \mathrm{P}(B)$, A, B independent;

$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B);$

$\mathrm{P}(A|B) = \frac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)};$ $\quad \mathrm{P}(B) = \mathrm{P}(B|A_1)\mathrm{P}(A_1) + \mathrm{P}(B|A_2)\mathrm{P}(A_2) + \cdots + \mathrm{P}(B|A_k)\mathrm{P}(A_k);$

$\mathrm{P}(A_j|B) = \frac{\mathrm{P}(B|A_j)\mathrm{P}(A_j)}{\mathrm{P}(B|A_1)\mathrm{P}(A_1)+\mathrm{P}(B|A_2)\mathrm{P}(A_2)+\cdots+\mathrm{P}(B|A_k)\mathrm{P}(A_k)};$

$F(x) = \int_{-\infty}^{x} f(y)\mathrm{d}y;$ $\quad \mathrm{E}[X] = \sum_{i=1}^{k} x_i \mathrm{P}(X = x_i),$ $\quad \mathrm{E}[X] = \int_{\mathbb{R}} x \cdot f(x)\mathrm{d}x;$

$\mathrm{Var}[X] = \mathrm{E}\left[(X - \mathrm{E}[X])^2\right] = \mathrm{E}[X^2] - \mathrm{E}[X]^2;$ $\quad \mathrm{SD}[X] = \sqrt{\mathrm{Var}[X]};$

$\mathrm{Cov}[X, Y] = \mathrm{E}[X \cdot Y] - \mathrm{E}[X] \cdot \mathrm{E}[Y];$ $\quad \mathrm{Corr}[X, Y] = \frac{\mathrm{Cov}[X,Y]}{\sqrt{\mathrm{Var}[X]}\cdot\sqrt{\mathrm{Var}[Y]}};$ $\quad Z = \frac{X-\mathrm{E}[X]}{\sqrt{\mathrm{Var}[X]}};$

Statistical learning:

$\widehat{f}(X) = \frac{1}{k}(Y_{i_1} + \cdots + Y_{i_k});$ $\quad \mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i^0 - \widehat{f}(x_i^0)\right)^2;$

$\mathrm{E}\left[\left(Y^0 - \widehat{f}(x^0)\right)^2\right] = \mathrm{Var}\left(\widehat{f}(x^0)\right) + \left[\mathrm{Bias}\left(\widehat{f}(x^0)\right)\right]^2 + \mathrm{Var}[\epsilon];$

$\mathrm{Bias}\left(\widehat{f}(x^0)\right) = \mathrm{E}\left[\widehat{f}(x^0)\right] - f(x^0);$

Linear regression:

$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x;$ $\quad \widehat{\beta}_1 = \frac{s_{y,n}}{s_{x,n}} \cdot r_{(x,y),n};$ $\quad \widehat{\beta}_0 = \overline{y}_n - \widehat{\beta}_1\overline{x}_n;$ $\quad e_i = y_i - \widehat{y}_i;$

$\mathrm{RSS} = \sum_{i=1}^{n} e_i^2;$ $\quad R^2 = \frac{\mathrm{TSS}-\mathrm{RSS}}{\mathrm{TSS}} = \frac{\sum_{i=1}^{n}(y_i-\overline{y}_n)^2 - \sum_{i=1}^{n}(y_i-\widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\overline{y}_n)^2} = \frac{s_{\widehat{y}}^2}{s_y^2};$

$R_{adj}^2 = 1 - \frac{\sum e_i^2/(n-k-1)}{\sum(y_i-\overline{y})^2/(n-1)};$

$\mathrm{RMSE} = \sqrt{\mathrm{MSE}};$ $\quad \mathrm{CV}_{(v)} = \frac{1}{v}\sum_{i=1}^{v}\mathrm{RMSE}_i;$ $\quad \mathrm{AIC} = \mathrm{constant} + n \cdot \ln\left(\frac{\mathrm{RSS}}{n}\right) + 2(k+1);$

$T_j(\mathbf{Y}, \mathbf{x}) = \frac{\widehat{\beta}_j(\mathbf{Y},\mathbf{x}) - \beta_{j,0}}{\mathrm{SE}_{\widehat{\beta}_j}(\mathbf{Y},\mathbf{x})} \sim t(n-k-1);$ $\quad \widehat{\beta}_j(\mathbf{Y}, \mathbf{x}) \pm t(n-k-1)_{1-\alpha/2} \cdot \widehat{\mathrm{SE}}_{\widehat{\beta}_j}(\mathbf{Y}, \mathbf{x});$

$\widehat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y};$ $\quad \widehat{\mathbf{y}} = \mathbf{X}\widehat{\beta} = \mathbf{H}\mathbf{y};$

$H_{ii} = \frac{\partial\widehat{y}_i}{\partial y_i};$ $\quad D_i = \frac{1}{(k+1)\widehat{\sigma}^2} \cdot (\widehat{\mathbf{y}} - \widehat{\mathbf{y}}_{(i)})^\top(\widehat{\mathbf{y}} - \widehat{\mathbf{y}}_{(i)});$

Logistic regression:

$\mathrm{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i};$ $\quad \mathrm{odds}(\{Y_i = 1\}|\mathbf{x}) = \frac{\mathrm{P}(\{Y_i=1\}|\mathbf{x})}{1-\mathrm{P}(\{Y_i=1\}|\mathbf{x})};$

$OR_j := \frac{\text{odds when } x_{i,j}=x+1}{\text{odds when } x_{i,j}=x};$ $\quad$ sensitivity $= \mathrm{P}(\text{Test} = + \mid \text{Condition} = +);$

specificity $= \mathrm{P}(\text{Test} = - \mid \text{Condition} = -);$ $\quad \beta = \mathrm{P}(\text{Test} = - \mid \text{Condition} = +);$

$\alpha = \mathrm{P}(\text{Test} = + \mid \text{Condition} = -);$ $\quad J = \text{sensitivity} + \text{specificity} - 1;$

$Z_j(\mathbf{Y}, \mathbf{x}) = \frac{\widehat{\beta}_j(\mathbf{Y},\mathbf{x}) - \beta_{j,0}}{\mathrm{SE}_{\widehat{\beta}_j}(\mathbf{Y},\mathbf{x})};$ $\quad \widehat{\beta}_j(\mathbf{Y}, \mathbf{x}) \pm z_{1-\alpha/2} \cdot \mathrm{SE}_{\widehat{\beta}_j}(\mathbf{Y}, \mathbf{x});$

$\mathrm{RR} = \frac{\mathrm{P}(\text{disease}|\text{exposed})}{\mathrm{P}(\text{disease}|\text{unexposed})};$

Foundations of Inference:

$\frac{\overline{X}_n-\mu}{\frac{\sigma}{\sqrt{n}}} \sim \mathcal{N}(0, 1);$ $\quad \frac{\overline{X}_n-\mu}{\sqrt{\frac{S_n^2(\mathbf{X})}{n}}} \sim t(n-1);$ $\quad \frac{n-1}{\sigma^2}S_n^2(\mathbf{X}) \sim \chi^2(n-1);$ $\quad \sum_{i=1}^{n} X_i \sim \mathrm{Bin}(n, \theta);$

$\left[\overline{X}_n - z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}, \overline{X}_n + z_{1-\alpha/2}\frac{\sigma_0}{\sqrt{n}}\right];$ $\quad \left[\frac{(n-1)S_n^2}{\chi^2_{1-\alpha/2}(n-1)}, \frac{(n-1)S_n^2}{\chi^2_{\alpha/2}(n-1)}\right];$

$T(\mathbf{x}) \pm z_{1-\alpha/2} \cdot \widehat{\mathrm{SE}}(T(\mathbf{X}))$ (based on CLT);

$\widehat{\theta}(\mathbf{X}) \pm z_{1-\alpha/2} \cdot \sqrt{\frac{\widehat{\theta}(\mathbf{X})(1-\widehat{\theta}(\mathbf{X}))}{n}};$ $\quad \widehat{\theta}(\mathbf{X}) \pm z_{1-\alpha/2} \cdot \widehat{\mathrm{SE}}(T(\mathbf{X}));$

$\mathrm{P}_{H_0}(\text{reject } H_0) \leq \alpha;$ $\quad \mathrm{P}_{H_A}(\text{reject } H_0) = 1 - \beta;$

$E_{ij} = \frac{\text{row } i \text{ total}}{\text{table total}} \cdot \text{column } j \text{ total};$ $\quad \chi^2 = \sum_{i=1}^{k}\sum_{j=1}^{\ell}\frac{(O_{ij}(\mathbf{X},\mathbf{Y})-E_{ij})^2}{E_{ij}};$

Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.